Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/19/2006
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Project Title:	Patterns of Energy Deposition by HZE Particles in Cellular Targets		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	End date changed to 7/31/2008 per JSC info update (10/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Toburen, Larry (East Carolina University)		
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	The interpretation of data from mechanistic studies on radiation biology requires detailed descriptions of the local patterns of energy deposition in the appropriate cellular or sub-cellular sites to explore the effects of radiation quality on biological response. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The calculations focus on interpretation of microscopic “event” or “dose” distributions appropriate to recently initiated studies of HZE radiation induced leukemia in animal and cellular models [NSCOR Program, Colorado State University, R. Ulrich, PI] and those encountered in the production of clustered damage in DNA [e.g., Brookhaven National Laboratory, B Sutherland, PI]. The progenitor cells for leukemia are haematopoietic stem cells that are located in clusters within marrow cavities having dimensions on the order of a few		

Task Description:	hundred microns. These cavities are surrounded by trabecular bone matrix, which forms a wall of several hundred microns thickness around the marrow. The dimensions and geometry of the bone marrow are too irregular for direct dosimetry measurements and dose-averaging methods are inappropriate for chronic low dose-rate (single particle) exposures. Detailed Monte Carlo charged particle track structure simulation will provide quantitative information on spatial patterns of energy deposition in cellular targets of dimensions as small as a few nanometers on a particle-by-particle basis; these distributions are the starting point to characterize the detailed nature of radiation-induced damage produced by high-LET radiation. The modeling approach will combine the capabilities of full event-by-event Monte Carlo track simulation codes with codes employing condensed history capabilities; the latter is more appropriate for the description of high-energy secondary electrons. The event-by-event ion code PROTRAC (which is part of the biophysical simulation tool PARTRAC) developed at GSF Neuherberg for protons and alpha particles will be extended for heavy charged particle (HZE) interactions using appropriate relativistic theory. Generated secondary electrons that traverse specific regions of interest, e.g., bone marrow cells or their nuclei, will be followed also event-by-event using ETRAC (the electron transport code of PARTRAC) whereas electron transport in adjoining tissues and the surrounding structural material will be followed by the more computer efficient condensed history general purpose MC transport code PENELOPE developed at the University of Barcelona. Appropriate interaction cross sections for a wide range of target materials will be added to the ETRAC and PROTRAC codes, specifically to explore the effects of bone-tissue interfaces on energy deposition in bone marrow cells. Comparison of the spatial pattern of energy deposited in volumes of interest obtained from models of the HZE particles with that obtained from low-LET radiation will provide a firm basis to interpret radiation quality effects on biological response and the mathematical structure developed is amenable to scaling to volumes appropriate to human dimensions for considerations of risk assessment. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The focus will be on Leukemia, i.e., on the bone/bone-marrow interface from realistic mouse or human dimensions down to single cells. Little is known on the interaction of HZE particles with the human body. HZE particles are part of the Galactic Cosmic Rays (GCR) which exist in deep space and cannot be shielded completely inside spacecrafts. The latency time of leukemia, 2-3 years, coincides more or less with the mission time for the planned expedition to Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Work is in progress in two directions. We have started work on the development of a HZE particle track structure module for the event-by-event Monte Carlo simulation code PARTRAC and we are developing a realistic bone/bone-marrow target model suitable for use with Monte Carlo simulations. Regarding the HZE particle transport we have developed an interaction model for bare (i.e., fully ionized) and sufficiently fast heavy ions by relating HZE interaction cross sections to ionization and excitation cross sections of protons by scaling laws. HZE inelastic interaction cross sections are obtained from proton cross sections by energy and charge scaling. Secondary electron emission angles are modeled as a superposition of soft collisions (dipole interaction) and hard or binary collisions. Proton interaction cross sections with liquid water have been derived earlier in the framework of a non-relativistic plane-wave Born approximation, the relativistic Bethe approximation and the dielectric response theory. The interaction model has been implemented into the Monte Carlo simulation code PARTRAC. Currently, HZE particles can be simulated from around 10 MeV/nucleon to 200 MeV/nucleon. Simulations above 200 MeV/nucleon lead to smaller energy losses or LET values than experimentally observed. Proton interaction cross sections have been reevaluated for some incident energies using a fully relativistic plane-wave Born approximation and the dielectric response theory. Calculations within these models are numerically delicate and computationally expensive. Some discrepancies between the relativistic Bethe and relativistic plane-wave Born approximation have been found. Further calculations are in progress. We also made progress in our goal to obtain interaction cross sections for HZE particles with calcium, a major component of bone. We collected a variety of data on optical properties and the dielectric response of (metallic) calcium in the literature. At the moment we are testing theoretical models to convert and calculate the dielectric response function of (metallic) calcium in the energy range from 0.01 eV to 100 GeV. We have tested our theoretical models with aluminum, where experimental and theoretical information is available. After our preliminary studies using the general purpose Monte Carlo code PENELOPE we have developed a realistic but practical bone/bone-marrow geometrical model for use with track structure simulations. The model is based on quadric geometry using the geometry package of PENELOPE and is based on the following assumptions: (1) The bone is spherical. (2) Bone marrow is enclosed in spheres, randomly distributed inside the bone. (3) Path length distributions in bone and bone-marrow follow experimental findings. (4) Known ratios of bone to bone-marrow volume and surface are used to determine the number of bone-marrow cavities. We are currently testing and verifying our bone/bone-marrow model in calculating path length distributions and absorbed fractions from emission of monoenegetic electrons from the bone and the bone-marrow (internal dosimetry). In general, our model reproduces the basic features of experimentally determined path length distributions. For further verification of our geometrical model we have calculated the absorbed fractions from internal emission of monoenergetic electrons from either the bone or the marrow. Comparisons with University of Florida’s model calculations show very good agreement in the absorbed fractions, both for the emission from bone and bone-marrow. Some discrepancies at higher incident electron energies are attributed to the fact that our model is an infinite model, while UF’s model has anatomically accurate external dimensions. The next step will be combining the geometrical model with external irradiation by electrons, alpha particles and HZE particles.
Bibliography Type:	Description: (Last Updated: 11/09/2018)
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